

# Rate Map

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## Menu: File > Rate Map

The Rate Map enables variable rate application by defining geographic zones with different target rates. It integrates with AgOpenGPS for GPS-referenced mapping.

## Prescriptions Tab

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Manages prescription shapefiles for the current field. The active prescription determines which zone rates are used during application.

| Control           | Action                                                                                                                                             |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Field             | Shows the field name assigned to the current job                                                                                                   |
| Prescription list | Lists all prescription files available for the current field. Select one to make it active. <b>Zones.shp</b> is the default and cannot be deleted. |
| New               | Create a new empty prescription file with a given name                                                                                             |
| Copy              | Copy the selected prescription to a new name. The copy becomes the active prescription.                                                            |
| Import            | Import a prescription shapefile from another location                                                                                              |
| Export            | Export the selected prescription shapefile to a chosen location                                                                                    |
| Delete            | Delete the selected prescription file. Zones.shp cannot be deleted.                                                                                |

**Note:** Selecting a prescription from the list immediately makes it the active prescription for the current job. The map reloads to show the selected zones.

## Zones Tab

Create and edit rate zones within the active prescription.

Create zones by clicking the  New button, then clicking at least **3 points** on the map to define the boundary.

- With **overlapping zones**, the **last zone added** takes priority.
- When variable rate is enabled but **no zone exists**, the **base rate** is used.

| Button                                                                                      | Action                                                                                                                     |
|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
|  New       | Create a new rate zone                                                                                                     |
|  Edit      | Edit zone properties (name, rate per product, color)                                                                       |
|  Delete    | Delete corner point (while creating) or delete zone (while editing)                                                        |
|  Cancel    | Cancel current edit                                                                                                        |
|  Save      | Save current edit                                                                                                          |
|  Zone List | Open the Zone List — view all zones with their areas, product rates (A–E), and colors; select/deselect zones; delete zones |

## Data Tab

| Control                                                                                    | Action                                                    |
|--------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| Record count                                                                               | Number of applied rate data points recorded               |
| A / B / C / D / E                                                                          | Select which product's applied rate to display on the map |
|  Record | Start/stop recording applied rates                        |
|  Delete | Delete all recorded applied rate data                     |

## VR Tab — Variable Rate Look-Ahead

Configures how far ahead of a zone boundary the application rate begins adjusting, so the rate matches the target as soon as the machine enters the new zone. Compensates for applicator response lag.

| Setting                              | Description                                                                                            |
|--------------------------------------|--------------------------------------------------------------------------------------------------------|
| <b>Look-Ahead Time (per product)</b> | Seconds before entering a zone boundary that rate adjustment begins. One value per product (up to 5).  |
| <b>Auto Tune</b>                     | RC monitors look-ahead performance and automatically adjusts the value to improve zone entry accuracy. |

## Files Tab

Import and manage supporting data files for the current field. Use the file selector dropdowns to choose which file is active for each overlay layer.

### Field Boundary

The field boundary defines the operated area and is used to clip automatically generated productivity zones. Select the active boundary from the **KML** dropdown. Import a boundary using the **Import Boundary** button — three file types are accepted:

- **AOG Field.kml** — the field KML from the AgOpenGPS Fields folder; the boundary polygon is extracted automatically.
- **AOG Boundary.txt** — AgOpenGPS boundary file; easting/northing coordinates are converted to lat/lon using the field origin.
- **Any KML (\*.kml)** — any KML file containing a polygon boundary.

**Tip:** Field files are typically found in:

- **AgOpenGPS:** Documents\AgOpenGPS\Fields\[Field Name]\ — contains Field.kml and Boundary.txt
- **TWOL:** Documents\TWOL\Fields\[Field Name]\ — contains Boundary.txt

### Yield, EC, and Elevation Data

| Control                   | Action                                                                                                                       |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Yield dropdown</b>     | Select the active yield data file for the overlay and zone creation                                                          |
| <b>Import Yield</b>       | Import a yield data CSV file. If the file contains elevation data, RC will offer to extract it to a separate elevation file. |
| <b>Delete Yield</b>       | Remove the selected yield data file                                                                                          |
| <b>EC dropdown</b>        | Select the active EC (electrical conductivity) data file                                                                     |
| <b>Import EC</b>          | Import an EC data CSV file (Lat, Lon, EC)                                                                                    |
| <b>Delete EC</b>          | Remove the selected EC data file                                                                                             |
| <b>Elevation dropdown</b> | Select the active elevation data file                                                                                        |
| <b>Import Elevation</b>   | Import an elevation data CSV file (Lat, Lon, Elevation)                                                                      |
| <b>Restore Elevation</b>  | Restore the previous elevation file from the automatic backup                                                                |
| <b>Delete Elevation</b>   | Remove the current elevation data file                                                                                       |

### Create Zones

The **Create Zones** button opens the productivity zone creation dialog. Zones are generated automatically from yield history, soil EC, elevation data, or any combination. Clear existing zones before generating if you want to replace them.

| Setting                        | Description                                                                                                                                                                                   |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Yield files (checklist)</b> | Check one or more yield data files to include. Multiple years are blended by their assigned weight.                                                                                           |
| <b>EC</b>                      | Include the active EC layer in zone generation                                                                                                                                                |
| <b>Elevation</b>               | Include the active elevation layer in zone generation                                                                                                                                         |
| <b>Weights</b>                 | Percentage weight for each active layer. All weights must sum to 100 before the Build button is enabled.                                                                                      |
| <b>Productivity levels</b>     | Number of productivity classes to create (e.g. 3 = Low / Med / High). Each class may produce more than one zone polygon if the field has non-contiguous areas of the same productivity level. |
| <b>Min zone size</b>           | Minimum zone area in hectares (or acres). Zone polygons below this size are discarded. Increase to reduce the number of small fragments.                                                      |
| <b>Build</b>                   | Generate zones using the selected layers and settings. The field boundary (set in the KML dropdown) is used to clip zones to the operated area.                                               |

**Tip:** Set the field boundary KML before generating zones. Without a boundary, zones are clipped to the bounding box of the data files, which may not match the actual field edge.

## Display Toggle Buttons (right panel)

These buttons highlight green when active. Click to toggle on or off.

| Button        | Action                              |
|---------------|-------------------------------------|
| Sat. View     | Toggle satellite background image   |
| Target Zones  | Toggle rate zone boundaries display |
| Applied Rates | Toggle applied rate colour overlay  |
| KML Files     | Toggle KML file overlay             |
| Yield Data    | Toggle yield data colour overlay    |
| EC            | Toggle soil EC colour overlay       |
| Elev.         | Toggle elevation colour overlay     |

## Bottom Controls

| Button                                                                                       | Action                                                             |
|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
|  Minimize | Shrink the map to a small window. Drag by title bar to reposition. |
|  Centre   | Centre the map on the current GPS position                         |
|  Help     | Open this help page                                                |
|  OK       | Close the Rate Map                                                 |